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# Professional Certificate in **Full-Stack ML Engineer** (FSWDML)

Duration: 180 Hours



## Objective:

Full-Stack ML Engineer teaches the student how to design, build, deploy, and maintain end-to-end machine learning systems, covering data engineering, MLOps, LLMs, cloud deployment, model optimization for edge/production environments, and responsible AI practices.

## What You'll Master

### End-to-End ML Systems:

Design, build, deploy, and maintain end-to-end Machine Learning systems.

### Data Engineering & Cloud:

Master data engineering, cloud deployment, and MLOps workflows.

### LLMs & Responsible AI:

Develop and fine-tune Large Language Models (LLMs) with Responsible AI practices.

### Model Optimization:

Optimize models for edge devices and high-throughput production environments.



## Curriculum Details:

**Module I:**  
Python  
Math for ML  
SQL  
FastAPI  
Software Engineering Principles

**Module II:**  
Cloud Storage  
Data Lakes  
Apache Spark  
Feature Engineering

**Module III:**  
Classical ML  
Deep Learning  
NLP  
Large Language Models (LLMs)  
MLflow

**Module IV:**  
MLOps  
CI/CD  
Infrastructure as Code (IaC)  
Cloud ML Platforms

**Module V:**  
Model Compression  
ONNX  
TensorRT  
Edge & Mobile Deployment

**Module VI:**  
ML System Design  
Capstone Project  
Ethics & Governance

## Skills You Graduate with

- ✓ Python Programming & Data Engineering
- ✓ Machine Learning & Deep Learning Architecture
- ✓ Large Language Models (LLMs) & Generative AI
- ✓ Model Optimization & Quantization
- ✓ MLOps & Cloud Deployment Pipelines
- ✓ Responsible AI & Production ML Systems



## Career Path

- ✓ Full-Stack ML Engineer
- ✓ MLOps Engineer
- ✓ Production Data Scientist
- ✓ NLP / LLM Engineer
- ✓ Edge ML / Embedded AI Engineer
- ✓ Computer Vision Engineer
- ✓ AI Founder / ML Consultant



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